# Periodic Properties of the Halogens

## Reminder: Dilution Lab Practical due soon!

Lab practicals will take place starting in this lab and continuing until your instructor's deadline. For more infor
see Instructions for Pipetting/Dilution for Lab Practical in the lab manual appendix.

## Goals

- To explore periodic trends among the halogens (Group 17 (VIIA) elements).
- To design and use qualitative identification techniques.
- Improve observation skills
- To begin to learn how to balance redox reactions

## Background

*Periodic Trends*

The Periodic Table of the Elements is so named because of the periodic variation of many important elementa
properties along a row (**Period**) or down a column (**Group** or **Family**) of the table. For example, atomic size
decreases from left to right along a Period as well as from bottom to top within a Group, as can be seen in Figu
below.

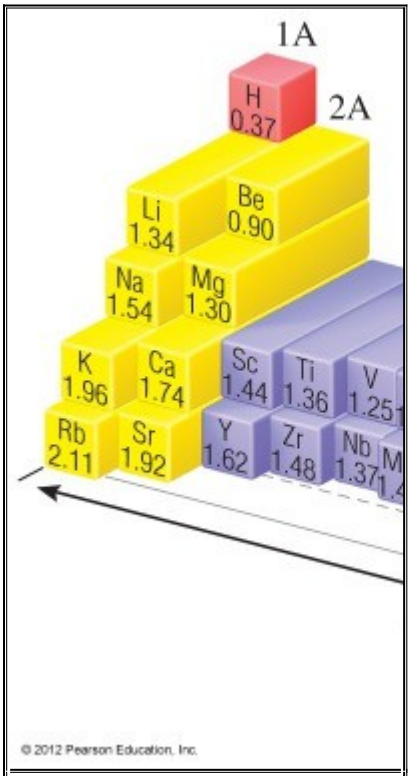



**Figure 1: Trends in bonding atomic radii for periods 1-5**
*(Ref: Brown et. al.; Chemistry: the*

*central science; 12th ed.; p. 255.)*

**Ionization energy, electronegativity, and electron affinity also vary in similar manners (see the sections i textbook).**

*Oxidation-Reduction Reactions*

The effects of these periodic variations in properties are often expressed in the relative chemical reactivities of members of a Period or Group. This experiment is designed to allow you to determine the relative reactivities (Cl, Br, and I) of the Group 17 elements (i.e the halogens) in **reduction - oxidation reactions** (referred to as **redox** reactions). Redox reactions involve the exchange of electrons. In the example below, 'X' represents a halogen (e.g. F, Cl, Br, or I) and '$Y^-$' denotes a halide ion. A reaction of halogen $X_2$with halide $Y^-$ would be as follows:

$$X_2 + 2\ Y^- \rightarrow 2\ X^- + Y_2$$

Redox reactions are best understood when written in terms of two half reactions.

| A Redox Reaction can be divided into two half reactions: | | |
|---|---|---|
| A half reaction that | With a reactant that undergoes | That reactant is defined as |
| Loses Electron(s) | Oxidation | Reducing agent |
| Gains Electron(s) | Reduction | Oxidizing Agent |

Viewed as a half reaction the **reduction** of $X_2$ would be:

$$X_2 + 2\ e^- \rightarrow 2\ X^- \qquad \textit{a gain of electrons}$$

And the **oxidation** half reaction of $Y^-$ would be:

$$2\ Y^- \rightarrow Y_2 + 2\ e^- \qquad \textit{a loss of electrons}$$

In this reaction, $X_2$ is being reduced; it is called the **oxidizing agent** because it is taking electrons away from Y causing $Y^-$ to be oxidized. Conversely, $Y^-$ is being oxidized; it is called the **reducing agent** because it is cau $X_2$ to be reduced.

Upon successful completion of the exercise, you will be able to arrange these halogens in order of their relativ reactivity as oxidizing agents. Likewise, you will be able to arrange the halides in order of their relative reactiv reducing agents.

*Qualitative Analysis*

A chemical reaction may be analyzed quantitatively or qualitatively. In **quantitative analysis**, the amounts of reactants and products are carefully measured and those data are used to make precise interpretations about the reaction. In **qualitative analysis**, changes in a reaction mixture are observed and that information is used to de whether or not a reaction ocurred. Qualitative changes include: the formation of a precipitate, the evolution of the release of energy as heat, the dissolution of a solid, and changes in the color or changes in clarity (clear vs. cloudy) of a solution. In this experiment, we will be observing color and clarity changes accompanying redox

reactions, i.e. we will be doing qualitative analysis.

*Spectator Ions and Net Ionic Equations*

Many ionic compounds dissociate in water to form solvated ions, as can be seen in Common Ion Chart from L
With rare exceptions, salts containing Group 1 elements are water soluble. For example, NaCl dissociates to fo
$_{(aq)}$ and $Cl^-_{(aq)}$, where (aq) means the ions are in an aqueous (water) solution. In addition to atomic ions, many
polyatomic ions exist that remain as a unit when dissociated, e.g. potassium permanganate, $KMnO_4$ in water fo
$_{(aq)}$ and $MnO_4^-{}_{(aq)}$. When we write a reaction, it can be written in terms of ionic compounds:

$$NaOH + HCl \rightarrow H_2O + NaCl$$

or in terms of solvated ions:

$$Na^+_{(aq)} + OH^-_{(aq)} + H^+_{(aq)} + Cl^-_{(aq)} \rightarrow H_2O + Na^+_{(aq)} + Cl^-_{(aq)}$$

You may notice that there are some species in the above equation that do not participate in the reaction. They s
as exactly the same ion/species on both sides of the equation. They are known as *spectator ions*. When a reacti
written in terms of solvated ions, the spectator ions should be removed, leaving the *net ionic equation*:

$$OH^-_{(aq)} + H^+_{(aq)} \rightarrow H_2O$$

This reaction is not a redox reaction. If it were, we might not be able to balance the reaction by inspection and
need to use the rules for balancing a redox reaction by the half-reaction method, as seen in ourAppendix. Note
will be balancing redox reactions using the half -reaction method in this lab and reporting any reactions that oc
lab in their final balanced net ionic reaction.

### Synopsis of the Experiment

In Part I of this experiment, you will develop a qualitative identification scheme which you can use to identify
presence of each halogen/halide used in the lab.

In Part II, you will mix each halide with each halogen to see which halogens are capable of oxidizing which ha
Using your identification scheme, you will be able to identify products if they form. In addition, you will com
each halogen with solid copper to see whether it can oxidize the copper. Using your results, you will be able t
conclusions about the relative oxidizing strengths of the halogens.

In Part III, you will mix each halide with $CuSO_4$ and with $KMnO_4$. Using your qualitative identification schem
will be able to identify products if the halides are successful as reducing agents. You will be able to draw con
about the relative reducing strengths of the halides.

## Preparation

### Reading Assignment:

- *Description of experiment*—see below.
- *The Group 7A Elements*—Brown *et al.*, Sec. 22.4, pp. 926–930.
- *Periodic Properties*—Brown *et al.*, Chapter 7, pp. 249-277.
- *Review Oxidation-Reduction Reactions*—Appendix & Brown *et al.*, Sec. 4.4, pp. 131-138.

### Prelab:

Click here for **Prelab worksheet**: pdf or Word format

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# Experiment

***To print instructions**, select the portion that you with to print, choose File/Print, and choose "selection" to pr printing the entire document.*

### Safety

***Aqueous chlorine and bromine may produce vapors that cause eye irritation. Make sure to keep the caps on these solutions when they are not in use.** Please rinse your eyes thoroughly with water if you experience the slightest eye irritation in lab.

***Copper sulfate solution is a skin and eye irritant.** Please rinse any affected area with copious amounts of water.

- Wear safety glasses!
- Do not pour solutions down the drain! Collect all halogen/halide wastes in the appropriately labeled dis containers provided.

### Materials and Equipment

- Test tubes (13 mm)
- Mineral oil
- Aqueous solutions of $Cl_2$, $Br_2$, and $I_2$
- 1 M aqueous solutions of NaCl, NaBr, and NaI
- 1 M $CuSO_4$
- 0.01 M $KMnO_4$ in 1 M $H_2SO_4$
- Cu turnings
- Parafilm

### Instructions

Students will work in groups of 2 unless otherwise specified by your instructor.

When preparing to perform an experiment, it is often advantageous to construct a chart or table in your notebo recording observations and results. This not only guides your work while you are conducting the experiment b makes review of the results easier after the work is complete. Please design an appropriate table for all 3 parts experiment. The data tables should make it easier for you to report your hypotheses (if applicable), observatio conclusions for every test tube you create in lab.

***Before you perform Part 1 - prepare three solutions for use in Part 2***

1. Obtain 3 test tubes.
2. In each test tube, place a small piece of copper turning (approximately 2 to 3 cm in length). Fill each ab full with a different aqueous halogen solution. Cover the test tubes with Parafilm and then allow them t for a minimum of one hour while you perform the rest of the experiment.

***Part 1. Preliminary Study - Making a Qualitative Identification Scheme*** (i.e.-creating "controls" for Part 2)

Before we can perform any reactions between various halides and halogens, we need a qualitative identificatio scheme which we can use to confirm the presence of the halogens and halides that will be reactants and produ our reactions. Begin by observing the provided aqueous halogen solutions and the aqueous halide solution in n oil (density of ~0.8g/mL) and without mineral oil (i.e.-in their own aqueous solutions where the aqueous solut density is ~1.0 g/mL).

In the elemental form, **halogens** ($X_2$) are neutral and preferentially dissolve in mineral oil because the molecul nonpolar. **Halides** ($X^-$) are negatively charged and readily dissolve in water because the ions are polar and sh very hesitant to dissolve in mineral oil.

Using test tubes, create your "controls" that show how all the halides and halogens appear in an aqueous envir and in the presence of mineral oil. Record your observations about each halogen and halides' appearence in the presence of water and mineral oil in a data table.. Make sure that you mix up any aqueous/mineral oil combina ensure that any halogen present finds it's way into the mineral oil.

Save these test tube solutions to use them as your "controls" when trying to identify which halogen/halide are in your reactions in Part 2.

*Part 1—Questions for thought:*

1. Are the solvents miscible (capable of being mixed)? If not, which solvent forms the top layer and which the lower layer?
2. What can you conclude about the solubilities of the halogens and halides in mineral oil? Explain the pr for one solvent or the other.
3. Do halides or halogens possess a unique color defining their presence? Which solvent accentuates the h presence in a solution?

***Part 2. Halogens as Oxidizing Agents***

When a halogen and halide from different elements are combined, (i.e.-$Br_2$ & $Cl^-$) two outcomes can occur: 1) nothing, or 2) the reaction will occur to produce the opposite halogen and halides (i.e.-$Br^-$ & $Cl_2$). You and you partner will be combining various halogens and halides and looking for indications of the presence of products halogens and halides from your starting materials) to determine whether or not a reaction occurred. In this way relative oxidizing strength of halogens can be determined.

Another way of testing the strength of halogens is to look for products when they are combined with metals. I halogen is introduced to elemental copper, $Cu_{(s)}$, a reaction may occur in which the elemental copper disoolves $Cu^{2+}_{(aq)}$ ion forms in the test tube (indicated by blue-green color).

**Prior** to performing any halogen/halide reactions in this part of the lab, prepare a data table that will allow you report the following information:
-- the specific halogen/halides you want to react in your test tubes
-- the hypothesis before the reaction takes place in the test tubes as to whether you think a reaction will take pl
-- the observations after mixing the halogen/halide reaction
-- the conclusion as to whether a reaction actually took place or not.

Now, proceed with your experiment combining various halogens and halides in new test tubes.

Combine the various halogens and halides in approximately equal volumes. Add ~1 cm of mineral oil to each and mix vigorously. Allow the contents of the test tubes to settle a moment and then record your observations conclusions in your data table.

After approximately one hour, record observations of your Cu/halogen test tubes. It can be difficult to observe reactions of Cu with halogens, even if you wait an hour, so take a look at the solutions prepared by your instru several hours prior to lab.

Use the "controls" from the Qualitative Identification Scheme from Part 1 to interpret the results Were your predictions correct? Remember that just because a color appears it does not mean that a reaction has occurred. reaction has occurred if you observe the disappearance of the reactants and/or evidence of the products.

*Please check with your instructor if your predictions do not come true in your experiment.*

*Part 2—Questions for thought:*

1. What physical change(s) did you observe which indicated that a reaction had occurred with the copper r
2. Write the balanced net ionic chemical equation for each reaction that occurred. Do not write an equatio reaction occurred. Note which chemical species is being oxidized and which is being reduced.
3. What statements can you make about the relative oxidizing strengths of the halogens? Support your stat by referring to experimental results. Arrange the halogens in order of increasing strengths as oxidizing a

***Part 3. Halides as Reducing Reagents***

One way of testing the reacting strength of a halide is to look for reaction products when it is combined with a

oxidizing agent that has a distinctive aqueous color that identifies its presence. The loss of the oxidizing agent' characteristic aqueous color indicates its disappearance upon reaction to product. **If an excess of halide is mi with a solution of copper sulfate ($CuSO_{4(aq)}$) or potassium permanganate ($KMnO_{4(aq)}$), the halide may act reducing agent to form solid copper halide ($CuX_{(s)}$) when excess halide is added to the reaction or manga dioxide ($MnO_{2(s)}$), respectively and it's corresponding halogen.**

Now that you are familiar with the general experimental approach from Part 2 for creating various reactions in tubes and how to distinguish which halogen is present in a reaction, design an experiment to investigate the rea of each of the aqueous halide solutions (using excess halide) with both copper (II) sulfate ($CuSO_{4(aq)}$) and acidi potassium permanganate ($KMnO_{4(aq)}$) solution. If a reaction occurs, the halide will produce its corresponding h and solid copper halide or solid manganese dioxide.

*Will mineral oil be helpful in identifying products for your reactions?* Generate a data table similar to Part 2's table to record your results. No predictions/hypotheses are required for this section. Be sure to develop an identification scheme for the aqueous oxidizing agents. Do you need a control for Part 3?

*Part 3—Questions for thought:*

1. Why were the copper ion and permanganate ions wise choices to serve as the oxidizing agents?
2. What statements can you make about the relative reducing strengths of the halides? Support your statem referring to experimental results. Arrange the halides in order of increasing strengths as reducing agents
3. Write the balanced net ionic chemical redox reaction for each reaction that occurred (net ionic means th spectator ions [ions that are not involved in the reaction] are not listed). Do not write an equation if no occurred. Note which chemical species is being oxidized and which is being reduced. See the Appendix guidelines and websites that will help you balance the redox reactions in this part of the experiment.

# Results

Click here for **Results worksheet**: pdf or Word format

*Question for thought:*

1. Explain the trends you discovered in this experiment based on the position of the halogens/halides in the Periodic Table. Use molecular and ionic size, ionization energy, electronegativity, and/or electron affini support your conclusions.

# For more information:

**Group 17 – The Halogens**:
http://www.chemsoc.org/viselements/pages/data/intro_groupvii_data.html
*Royal Society of Chemistry web site which has a visual elements section and other chem. stuff*

**Introduction to the Halogens**:

http://www.wpbschoolhouse.btinternet.co.uk/page03/The_Halogens.htm
*"Doc Brown Chemistry Clinic"  by a chemistry teacher in the UK*
*very colorful, with some cute graphics (can "cute" and "chemistry" be used on the same page?!)*

**Redox Balancing Practice Problems**:
http://wc.pima.edu/~skolchens/C152OL/Ch21/REDOX.htm
http://www.chemistrycoach.com/Redox.htm (Has solutions too!)

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